State of the art report - NeRF (Neural Radiance Fields) and similar techniques

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Abstract

Neural Radiance Fields (NeRF) have revolutionized 3D representation and novel view synthesis by using neural networks to model complex scenes from 2D images. In this report, we will provide an overview of the original NeRF method, its variants and similar techniques, discussing their strengths and limitations and future directions.

1. Introduction

Traditional 3D reconstruction methods, such as SfM (Structure from Motion) [Snavely et al., 2008] and MVS (Multi view stereo) [Seitz et al., 2006] construct discrete representations (points cloud, mesh) that often lack photo realism. NeRF (Neural Radiance Fields) addresses this limitation by using a neural network to model the scene as a continuous volumetric function. NeRF functions with a neural radiance field meaning a fully connected deep network (or multi-layer perceptron, MLP) that maps 3D coordinates and the corresponding viewing directions to volume density and RGB colors. This allows for photorealistic rendering from novel viewpoints, better modeling of complex lighting or surfaces, textures and end-to-end optimization from 2D images without 3D modeling needed. NeRF is the basis of many research projects in computer vision with applications in fields such as virtual reality, gaming, robotics and many more.

2. Background

Originally volume rendering is a technique used for generating 2D images from 3D volumetric data. Rays are cast from a virtual camera through the scene (ray marching), the density and color is evaluated at certain points along each ray. By combining the collected samples, we obtain the final pixel color. However, classical volume rendering methods such as splatting or shear-warp are computationally expensive and require explicit 3D models.

Deep learning has been applied to graphics for things such as image based rendering, 3D reconstruction, or differentiable rendering to enable gradient-based optimization [Loper et al., 2014]. Differentiable rendering allows neural networks to be trained directly on images by backpropagating gradients through the rendering process. This is key for NeRF, as it enables optimization via photometric loss.

3.Techniques

3.1. Original NeRF

NeRF represents a scene as a continuous 5D function with as input (x,y,z,θ,ϕ), where (x,y,z) is the 3D location of a point and (θ,ϕ) the viewing direction from which it is captured. The output of the function is (σ, c) where σ is the volume density of the point (how much it

blocks the light, its opacity) and c, the RGB color of the point. The function is parametrized by a MLP (multi-layer perceptron) with ReLu activation (see figure 1.).

To model high frequency details, the technique uses positional encoding, which allows the input coordinates to be mapped to higher dimensional spaces with sinusoidal functions, enabling the MLP to render complex scene geometry and appearance. For each pixel in the rendered image, like explained above, a ray is cast through the camera on the scene. The points are sampled along the ray, and with the MLP the density and color of each pixel is predicted. The contribution of all sampled points is combined using volumetric rendering equations that are detailed in the source article to produce the final pixel. This process is differentiable, allowing the network to be optimized end-to-end using only 2D image supervision.

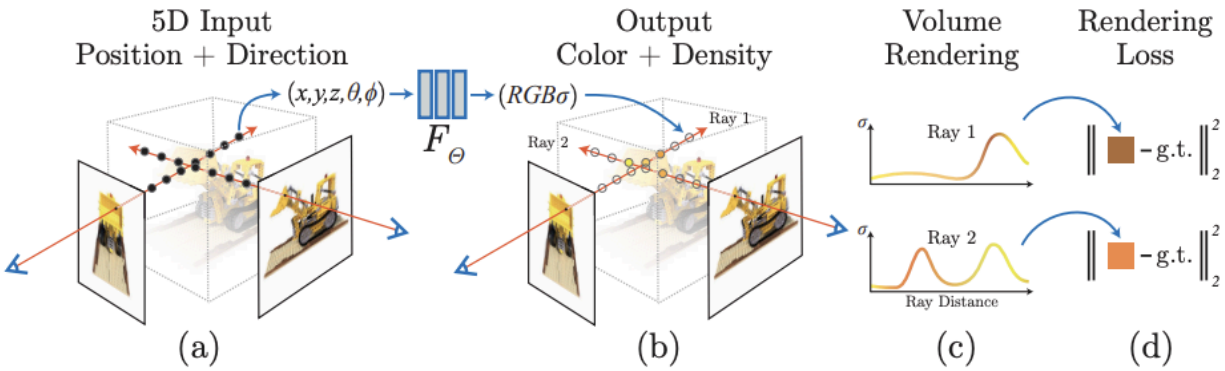

Figure 1 - overview of our neural radiance field scene representation and differentiable rendering procedure [Mildenhall et al.].

During training, the loss function used is the Photometric Loss function (MSE) between the rendered and real images. The network optimizes through SDG (Stochastic Gradient Descent) on batches of rays. The input of the model are posed images, and the poses (positions of the camera) can be determined via COLMAP [Schönberger and Frahm, 2018]. This technique manages to achieve state-of-the-art view synthesis of complex scenes, while also capturing view-dependent effects (reflections, material textures, lighting, etc…).

3.2. Variants

There are several other methods that have been developed based on the original NeRF technique. For example, MIP-NeRF [Barron et al., 2021] addresses a limitation of the original paper: since it samples a single ray per pixel, the renderings might be aliased or excessively blurred. MIP-NeRF addresses this by representing the scenes as mipmaps (conical fostrums, cones) instead of points, to achieve better precision of image rendering.

In another improvement of NeRF Müller et al., 2022 propose a solution, Instant-NeRF, that speeds up the training process. Instead of an MLP it uses a hash grid for faster lookups, which allows it to achieve real-time rendering. Several other techniques complexifying the existing method of NeRF such as Dynamic NeRF [Pumarola et al., 2021] which adds time as

an input for time varying scenes, or GAN-NeRF [Schwarz et al., 2020] combining the NeRF with a GAN discriminator to improve the realism of the scene based on its outputs.

4. Challenges and future directions

The first challenge in the context of NeRF is the need for high computational resources both for training and rendering, using solutions such as Instant-NeRF above help address this issue by using less complex networks for faster results. Furthermore, since it is an MLP-based architecture, it is memory-bound meaning the scalability is limited. Balancing quality of the rendering with resource usage and speed represents one of the main challenges for improving NeRF.

NeRF is usually scene-specific and needs retraining to adapt to new scenes. It also requires a large number of input images: 50 to 100 views to construct a scene. This complicates the generalization of the technique, that would work across diverse scenes, and also require less input.

Other areas of improvement for NeRF include addressing the idea of dynamic scene capturing, including a time dimension. Making the method more accessible is also key, as NeRF currently requires controlled captures, with known pose views and specific cameras work, also only taking static scenes. NeRF could also include and work with different modalities, such as audio or LiDAR [Pan et al., 2022; Gao et al., 2022], to make for even more realistic scene rendering and enlarge the applications field.

5. Conclusion

NeRF has revolutionized novel view synthesis by introducing a continuous, neural representation of 3D scenes. While the original method set a new standard for photo-realism, recent advances have addressed many of its limitations, including speed (Instant NGP), generalization (PixelNeRF), dynamics (Dynamic-NeRF). However, challenges remain in real-time deployment, multi-modal integration, and theoretical understanding. Future work will likely focus on interactive applications, cross-modal fusion, and efficient, generalizable NeRF models, further expanding its impact in both research and industry.

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